

MS5612 - Projects Proposal

Ajith, Hardik, Lokesh, Vikrant

August 7, 2026

1 Buffet Options in Restaurants

Many restaurants offer buffet services in their menus, these are generally more expensive than the regular menu options available. Buffet options for a customer can be modelled as a vanilla call option, wherein the customers only pay a fixed amount (premium) irrespective of the quantity and items they consume.

Interestingly, buffets options often exclude some items and those items are only served once (for example, the blue welcome drink in Rajdhani, Phoenix), and the customers have to pay for them separately, This can be modelled as classic “security + call option”, albeit a different security.

We plan to understand how the buffet options are priced by gathering primary data from nearby restaurants that offer this service.

2 Spinny’s Buyback Assurance

Spinny, an online used-car marketplace, offers buyers a 90-day buyback assurance — after purchasing a car, the customer can sell it back to Spinny within 90 days at a guaranteed price fixed at the time of sale.

This can be modelled as a put option held by the buyer: they have the right, but not the obligation, to sell the car back at a pre-agreed price if its resale value falls or they simply change their mind. Spinny effectively writes this option and recovers its cost through the gap between the purchase price and the buyback price.

We plan to understand how this buyback is priced by gathering primary data from Spinny hub staff and a short survey of used-car buyers on how they value this safety net.

3 Expansion of Real Estate Projects

Real estate developers in Chennai require a substantial amount of capital. They also face large uncertainty regarding tenant occupation within their projects. In this context, we consider the Option to Phase & Expand. Instead of fully building a facility, developers only construct an initial footprint at first. They have a call option to develop more construction phases only if rental income and demand is sufficient.

Since the internal financial data (e.g. firm hurdle rate, lease rates) of the developers are likely confidential, interviewing them directly on this topic will be difficult. Instead, we can interview local industry experts.